

Problem

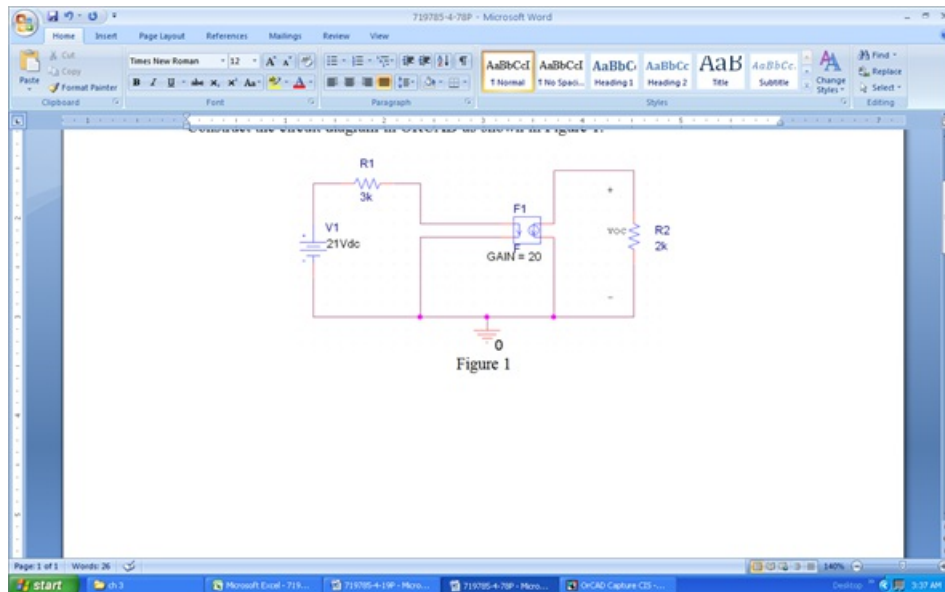
Use *PSpice* or *MultiSim* to solve Prob. 4.52.

Step-by-step solution

Step 1 of 16

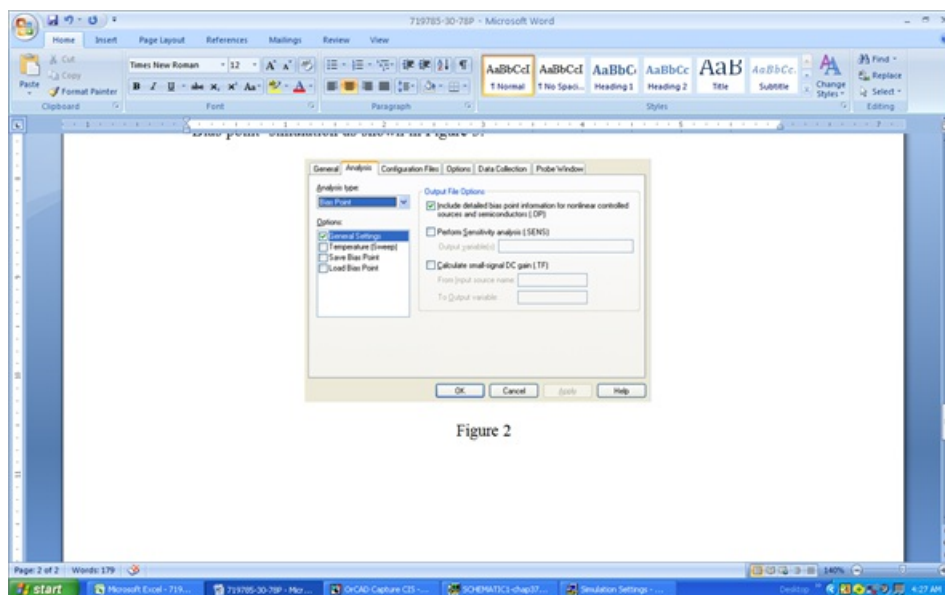
Refer to Figure 4.118 in the text book.

Construct the circuit diagram in ORCAD as shown in Figure 1. Keep the output terminals open.



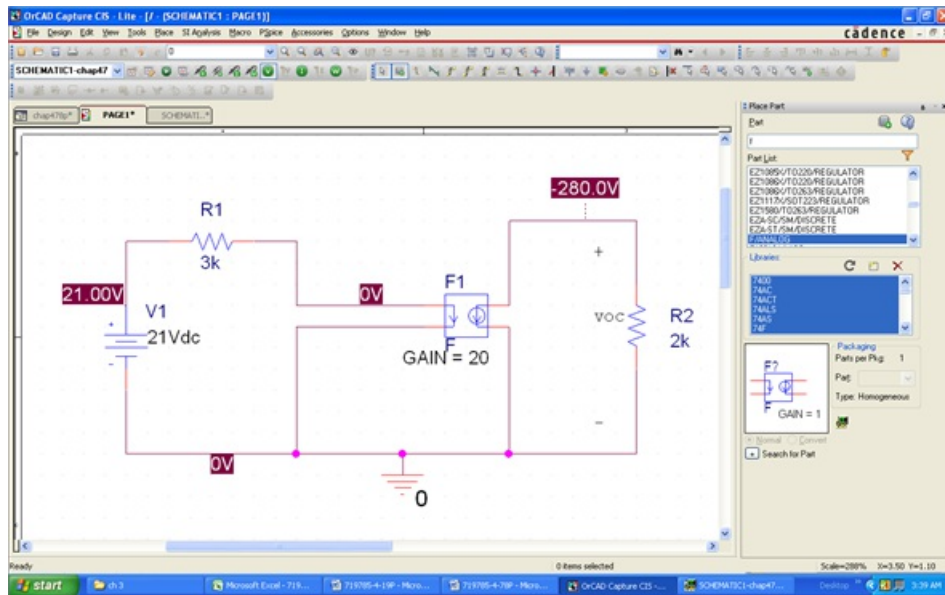
Step 2 of 16

Go to Pspice and create a new simulation profile. Edit the simulations profile and select 'Bias point' simulation as shown in Figure 2. Click on ok.



Step 3 of 16

Click on voltage bias simulation display. Press F11 to simulate the circuit.

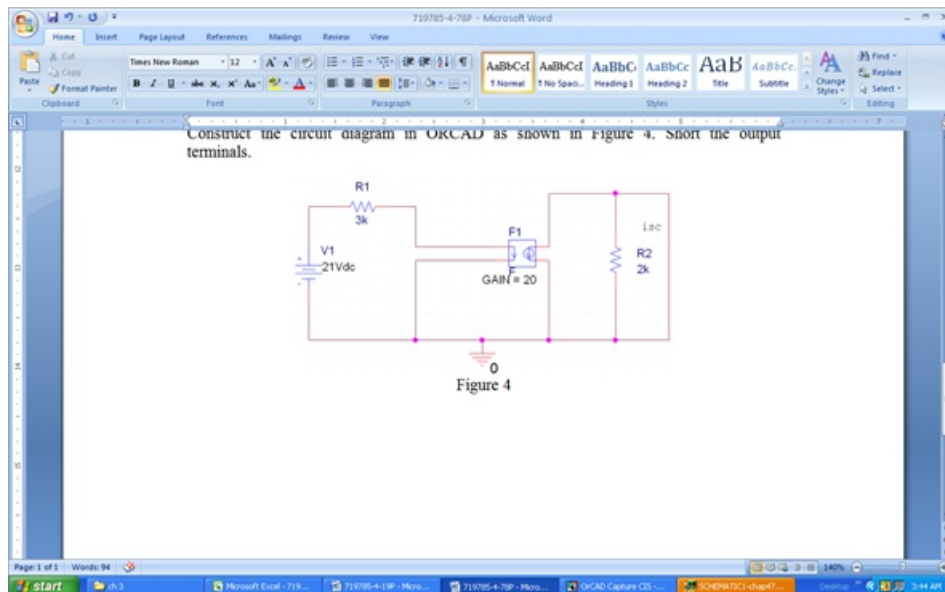


Step 4 of 16

Therefore, the value of the open circuit voltage is -280 V .

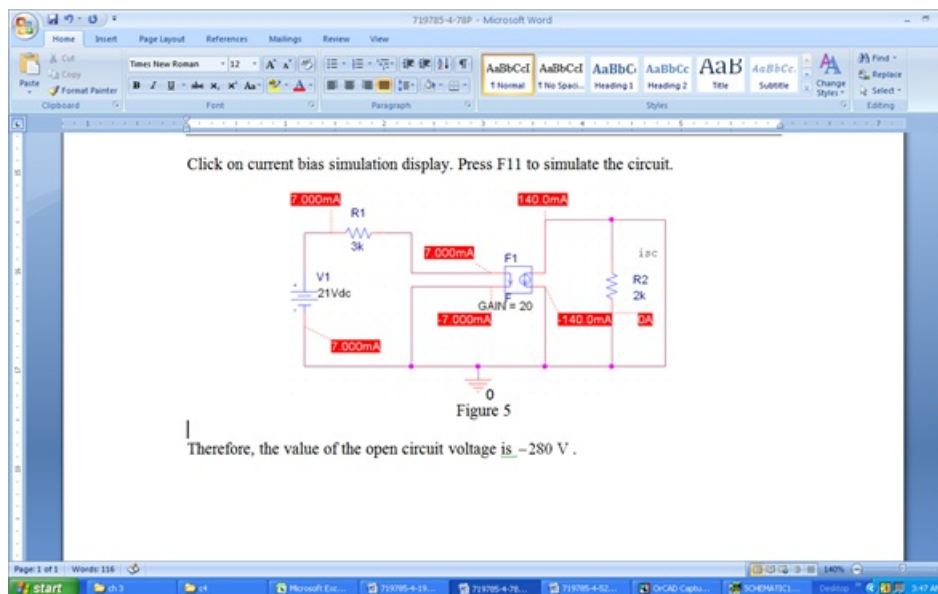
Step 5 of 16

Construct the circuit diagram in ORCAD as shown in Figure 4. Short the output terminals.



Step 6 of 16

Click on current bias simulation display. Press F11 to simulate the circuit.



Therefore, the value of the short circuit current is -140 mA .

Step 7 of 16

Calculate the value of the value of the Thevenin's resistance.

$$R_{Th} = \frac{v_{oc}}{i_{sc}}$$

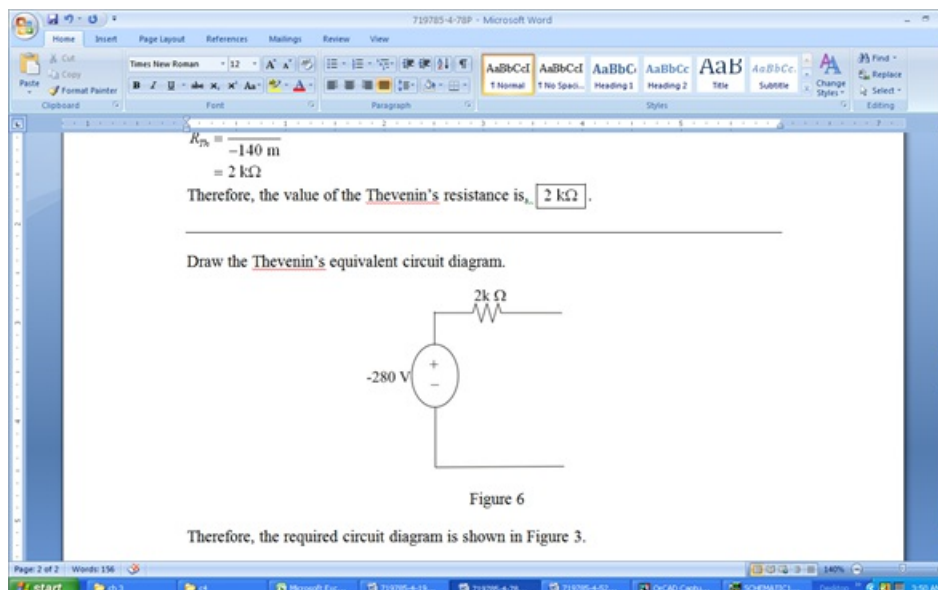
Substitute -140 mA for i_{sc} and -280 V for v_{oc} .

$$R_{Th} = \frac{-280}{-140 \text{ m}} = 2 \text{ k}\Omega$$

Therefore, the value of the Thevenin's resistance is, $2 \text{ k}\Omega$.

Step 8 of 16

Draw the Thevenin's equivalent circuit diagram.

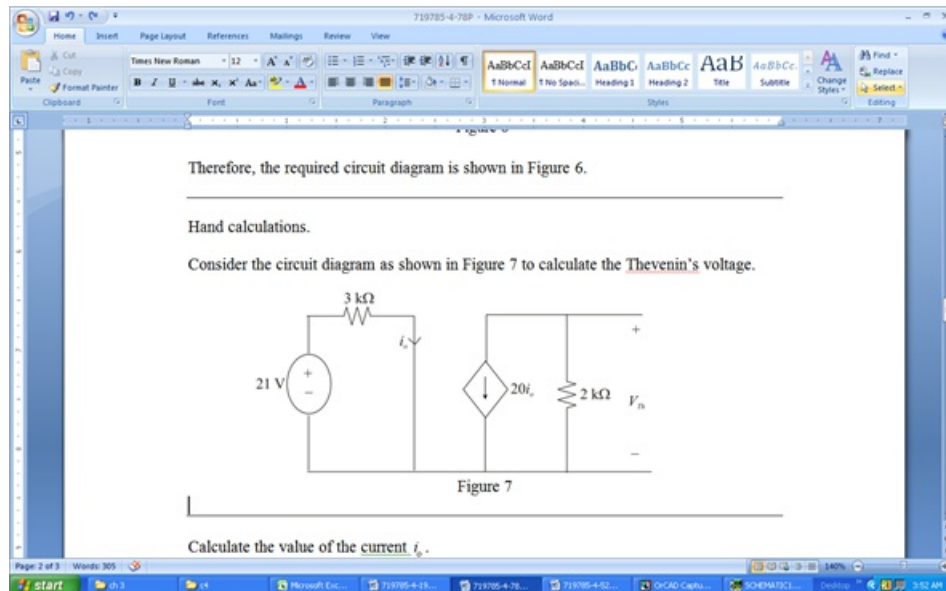


Therefore, the required circuit diagram is shown in Figure 6.

Step 9 of 16

Hand calculations.

Consider the circuit diagram as shown in Figure 7 to calculate the Thevenin's voltage.



Step 10 of 16

Calculate the value of the current i_o .

$$\begin{aligned} i_o &= \frac{21}{3 \text{ k}} \\ &= 7 \text{ mA} \end{aligned}$$

Therefore, the value of the current i_o is, 7 mA.

Step 11 of 16

Calculate the value of the Thevenin's voltage V_{Th} .

$$V_{Th} = (-20i_o)(2 \text{ k})$$

Substitute 7 mA for i_o .

$$\begin{aligned} V_{Th} &= (-20)(7 \text{ m})(2 \text{ k}) \\ &= -280 \text{ V} \end{aligned}$$

Therefore, the value of the Thevenin's voltage V_{Th} is, -280 V.

Step 12 of 16

Consider the circuit diagram shown in Figure 8 to calculate the short circuit current.

Step 13 of 16

No current passes through $2 \text{ k}\Omega$ resistor since it is shorted.

Calculate the value of the current i_o .

$$\begin{aligned} i_o &= \frac{21}{3 \text{ k}} \\ &= 7 \text{ mA} \end{aligned}$$

Therefore, the value of the current i_o is, 7 mA .

Step 14 of 16

Calculate the value of the Norton's current I_N .

$$I_N = -20i_o$$

Substitute 7 mA for i_o .

$$\begin{aligned} I_N &= (-20)(7 \text{ m}) \\ &= -140 \text{ mA} \end{aligned}$$

Therefore, the value of the Norton's current I_N is, -140 mA .

Step 15 of 16

Calculate the value of the value of the Thevenin's resistance.

$$R_{Th} = \frac{V_{Th}}{I_N}$$

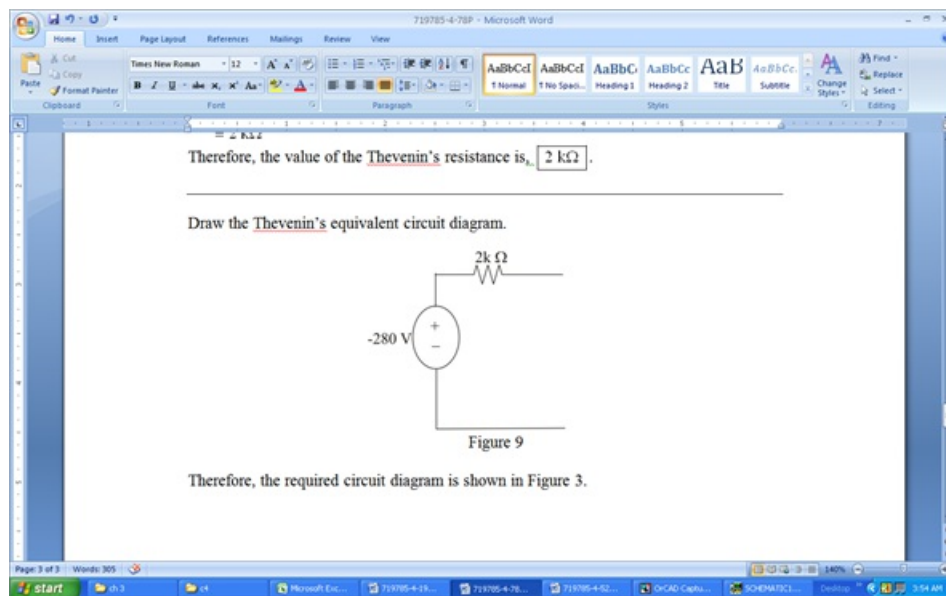
Substitute -140 mA for I_N and -280 V for V_{Th} .

$$\begin{aligned} R_{Th} &= \frac{-280}{-140 \text{ m}} \\ &= 2 \text{ k}\Omega \end{aligned}$$

Therefore, the value of the Thevenin's resistance is, $2 \text{ k}\Omega$.

Step 16 of 16

Draw the Thevenin's equivalent circuit diagram.



Therefore, the required circuit diagram is shown in Figure 9.